Data Consistency: Inserting with Foreign Keys

Lecture Notes · Game DB Junior

In this lesson

- What **data consistency** means
- The **three cases** when we insert a photo with a foreign key
- The **foreign key constraint** error and the special value **NULL**

1. What is data consistency?

Data consistency is the ability of our database to make sure that the data we insert and work with **makes sense**, and that the references between tables really **line up**.

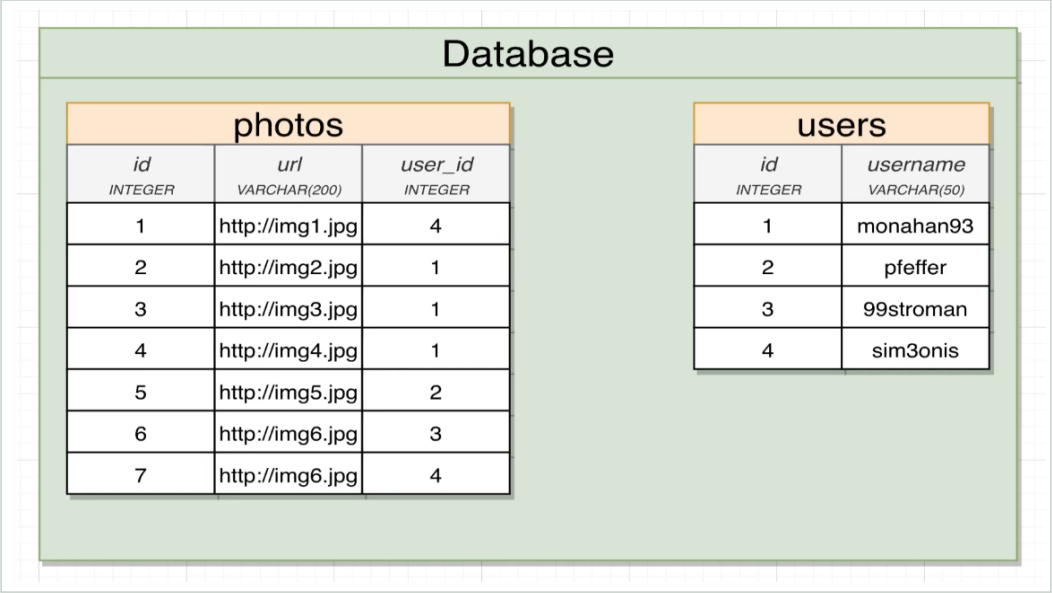


Figure 1 — The photos and users tables (users have ids 1 to 4)

2. Three cases when we insert a photo

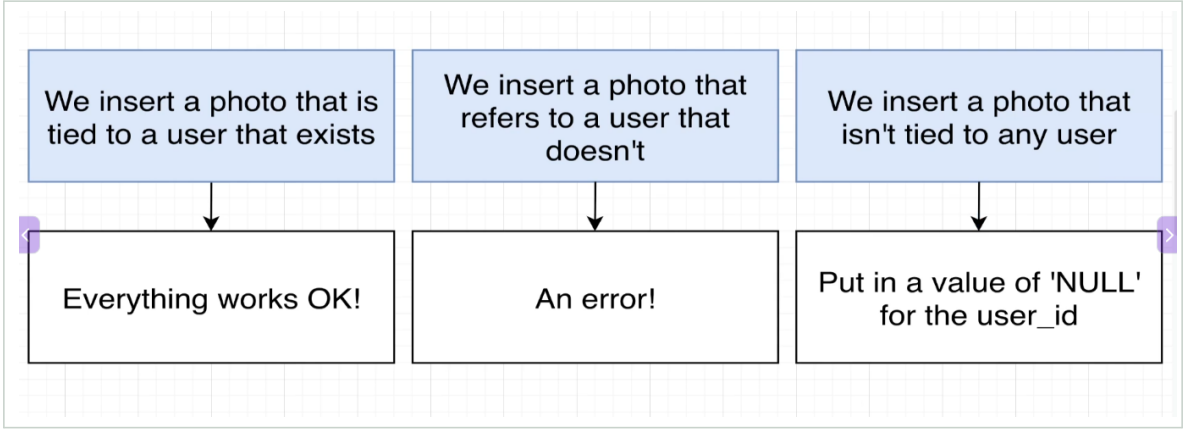


Figure 2 — The three cases

Case	What happens
1. The photo is tied to a user that **exists**	Everything works OK
2. The photo refers to a user that **does not exist**	An error
3. The photo is **not tied to any user**	Put in a value of NULL for user_id

Case 1: the user exists

We give a user_id that is in the users table: 1, 2, 3 or 4. This is like all our inserts so far. The photo is inserted with no problem, and we can query the photos of that user.

Case 2: the user does not exist

We only have users with ids 1 to 4. What if we try to tie a photo to user **9999**? This does not make sense in any way, so we get an **error**.

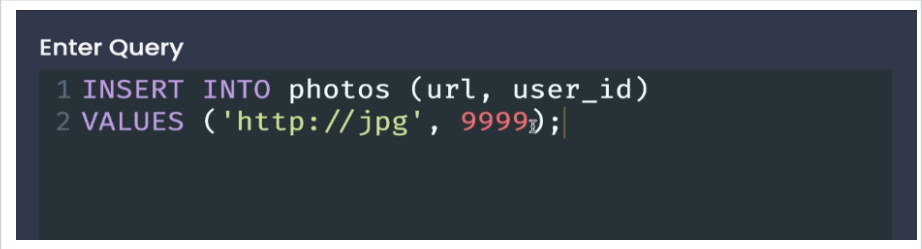


Figure 3 — Inserting a photo for user 9999

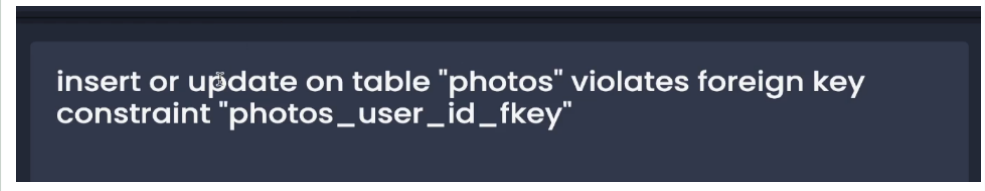


Figure 4 — Error: violates foreign key constraint

- The error says the insert on photos **violates a foreign key constraint** (photos_user_id_fkey).
- **Foreign key constraint**: Postgres makes sure the foreign key points at a record that **really exists** in the users table. Here there is no match, so we get an error.
- This is a **very good feature**. It stops us from putting data into the database that makes no sense.

Case 3: no user at all

Some photos may not be tied to any user. Example: an **image of the day** that shows up when a user logs in. For this, we put **NULL** as the user_id.

> **NULL** is a very special value in SQL and Postgres. It means **there is no value here**: nothing.

Inserting a photo with user_id NULL works.



Figure 5 — user_id is NULL

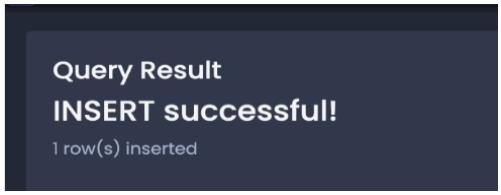


Figure 6 — INSERT successful

With **SELECT * FROM photos;** we see the last photo has **null** in user_id. This means no user is tied to this photo.

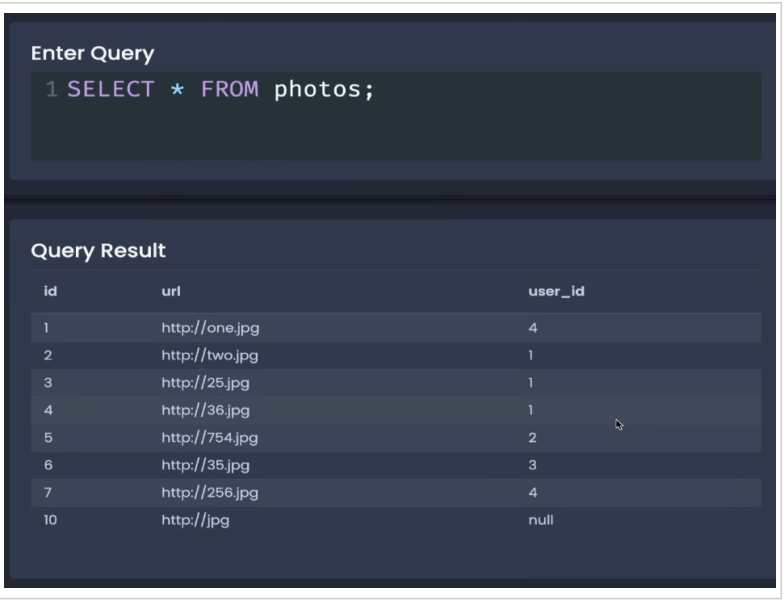


Figure 7 — The last photo has user_id null

3. Summary

Idea	Remember
Data consistency	The data makes sense and the references between tables line up
User exists	The insert works
User does not exist	Error: violates foreign key constraint
No user	Use NULL for the foreign key
NULL	Means there is no value

This was only about **inserting**. **Next lesson:** what happens with foreign keys when we **update** or **delete** a record.

Sources: Lecture transcript and lecture images (Figures 1–7)